ADSDB P1 / UPC

EcoSentinel

Where does the answer come from?

Mario Rodrigo · Sambhav · Wassim



Photo: lynxexsitu.es / CC BY 3.0 es

A photograph starts the investigation

“What can help me understand this photograph?”

Related images
Recordings
Source descriptions



Education and biodiversity exploration

Photo: Alexkom000 / CC BY 4.0

Three modalities, one reference dataset

198

accepted assets
10 Iberian wildlife species

100 Images / Wikimedia

34 Audio / Xeno-canto

64 Text / Wikipedia + GBIF

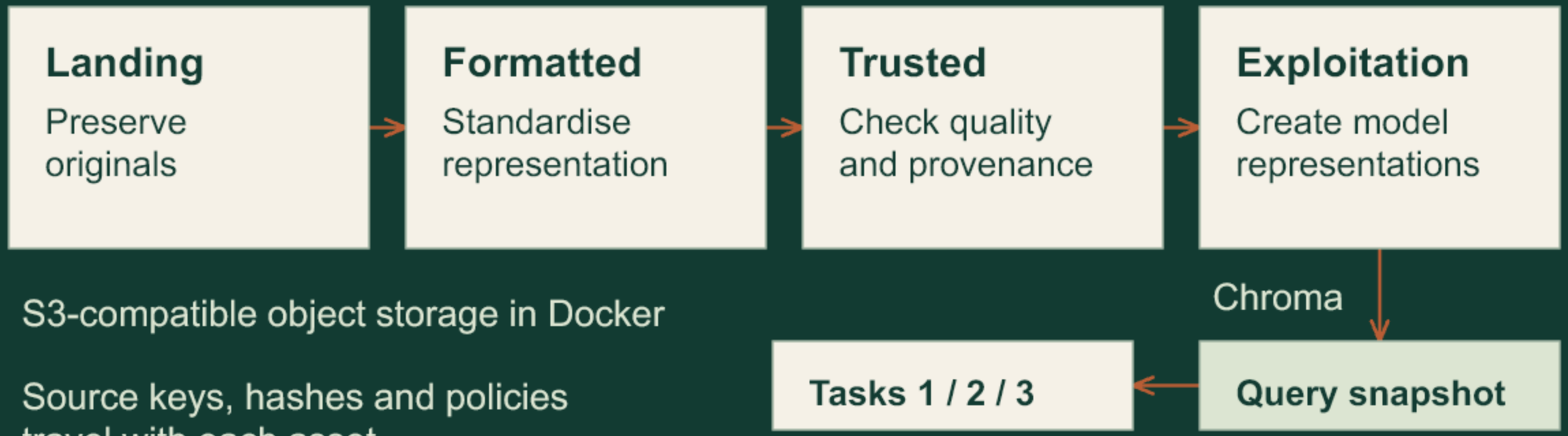
Uneven availability remains visible: no audio for the ocellated lizard

The EcoSentinel pipeline

Optional acquisition or local files / Security Gate

16-step orchestrator

Lock, timeouts and step reports



S3-compatible object storage in Docker

Source keys, hashes and policies travel with each asset

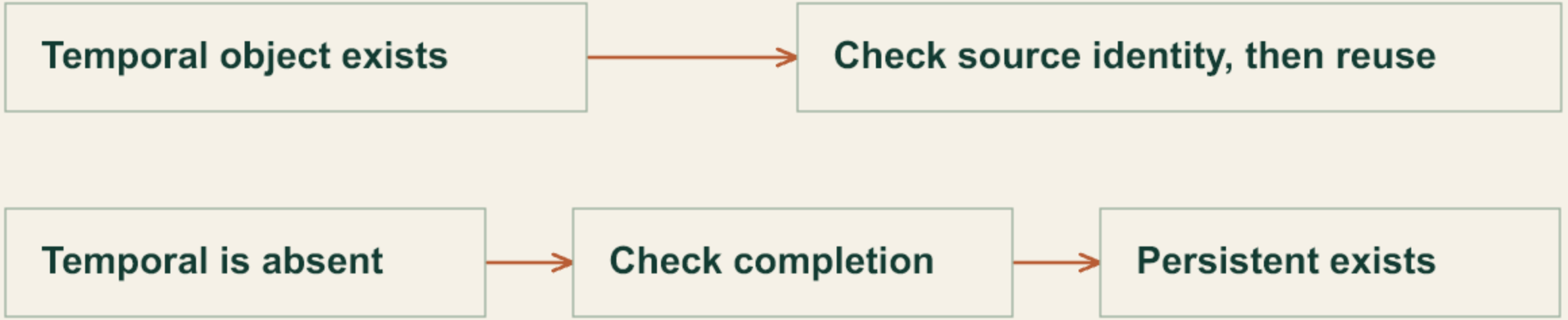
Hashes and embeddings

	SHA-256 hash	Embedding
Question	Did the bytes change?	Which assets are similar?
Output	64 hexadecimal digits	384 or 512 numbers
EcoSentinel use	Identity and integrity	Similarity and retrieval

Re-encode a photo: a different hash, potentially similar embeddings

Landing identity and completion records

temporal_landing/<SHA-256[:16]>__<filename>



Matching identity + recorded Persistent target: skip
No completion record: ingest as new

Promotion verifies the copy before completion. Missing targets or conflicts fail.

Landing: four identity cases

Same source metadata; the first file is already recorded

Already seen	Incoming file	Key change	Action
 lynx.jpg hash de7879...	 lynx.jpg hash de7879...	None	SKIP
 lynx.jpg hash de7879...	 renamed.jpg hash de7879...	Name	INGEST
 lynx.jpg hash de7879...	 lynx.jpg hash 3e983a...	Hash	INGEST
 lynx.jpg hash de7879...	 renamed.jpg hash 3e983a...	Both	INGEST

Technical ingestion reuse; Trusted still checks image content

Formatted choices for wildlife data

Asset	Canonical format	Why it fits this pipeline
Image	RGB JPEG, quality 95	Compact photographic representation Keep original width and height
Audio	FLAC PCM16	Compressed PCM for wildlife audio Keep rate, channels and duration
Text	UTF-8, LF, NFC	Consistent multilingual decoding Readable input for quality and retrieval

Real exception: 6 multiframe sources contribute their primary frame

Trade-offs: JPEG loss, PCM16 quantisation, extracted layout. Originals stay in Persistent.

Trusted: rejection or warning?

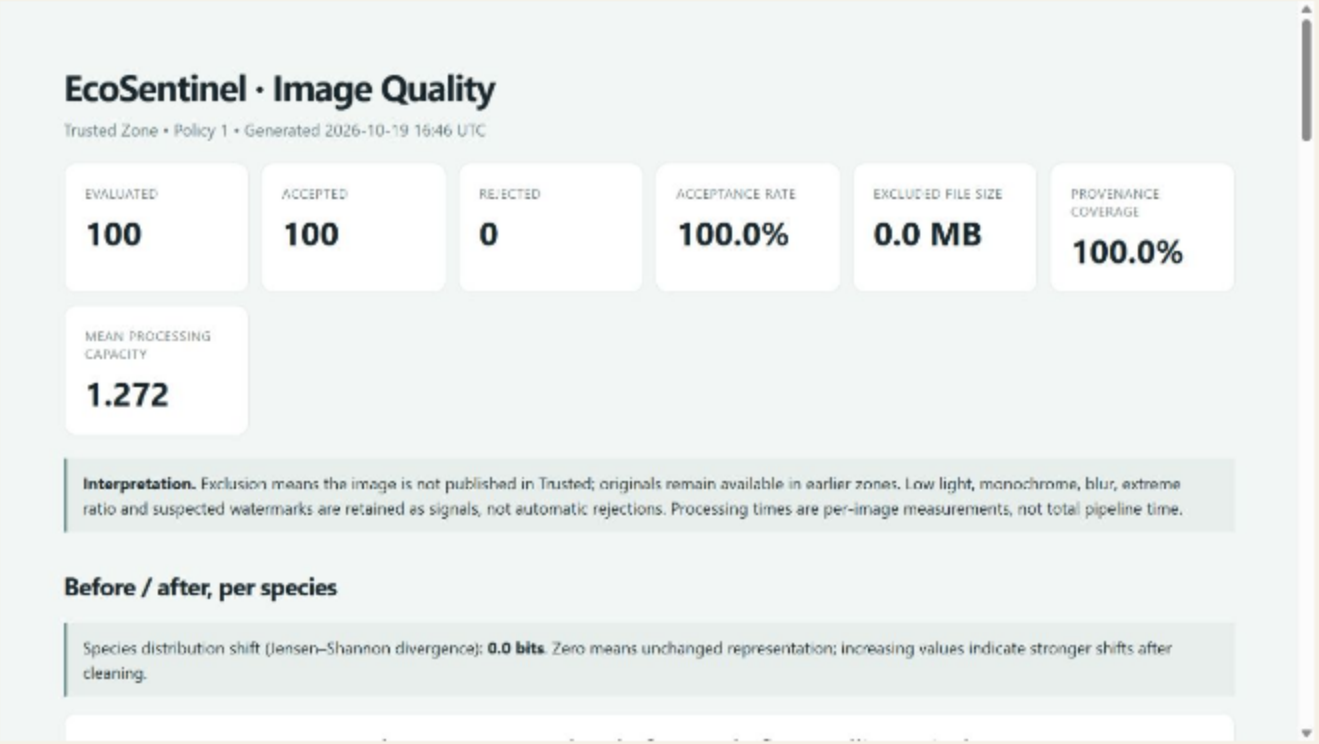
Observation	Policy
Invalid decoding	Reject with a reason
Confirmed image duplicate	Exclude the later matching image
Watermark / low-energy audio	Retain with a warning

Image budget: 20 MB / 40 MP / minimum 128 × 128

Audio: 200 MB / text: 4 MB / preserve previously accepted bytes after later failures

Quality decisions leave an audit trail

Per-asset manifests / HTML + CSV + JSON + plots



EcoSentinel · Image Quality

Trusted Zone • Policy 1 • Generated 2026-10-19 16:46 UTC

EVALUATED	ACCEPTED	REJECTED	ACCEPTANCE RATE	EXCLUDED FILE SIZE	PROVENANCE COVERAGE
100	100	0	100.0%	0.0 MB	100.0%

MEAN PROCESSING CAPACITY
1.272

Interpretation. Exclusion means the image is not published in Trusted; originals remain available in earlier zones. Low light, monochrome, blur, extreme ratio and suspected watermarks are retained as signals, not automatic rejections. Processing times are per-image measurements, not total pipeline time.

Before / after, per species

Species distribution shift (Jensen–Shannon divergence): **0.0 bits**. Zero means unchanged representation; increasing values indicate stronger shifts after cleaning.

100%

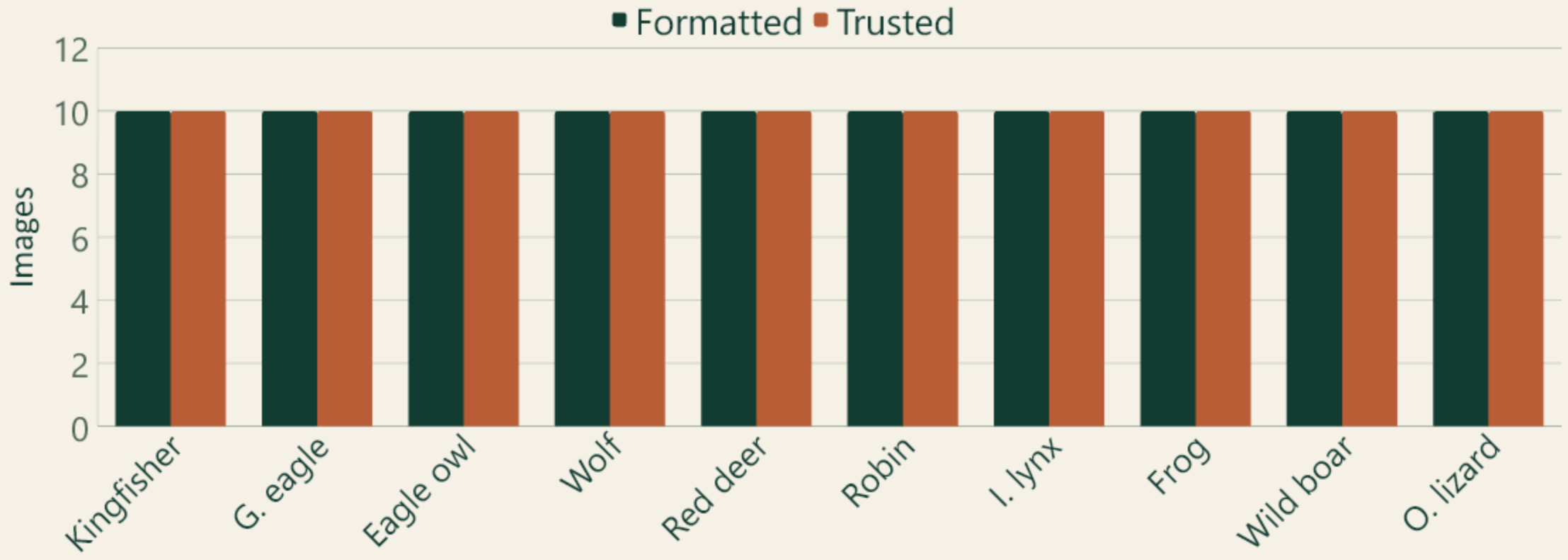
Provenance fields complete

0.0 bits

Species distribution shift

Real image report / 100 retained / 32 possible watermark flags / 19 Oct 2026

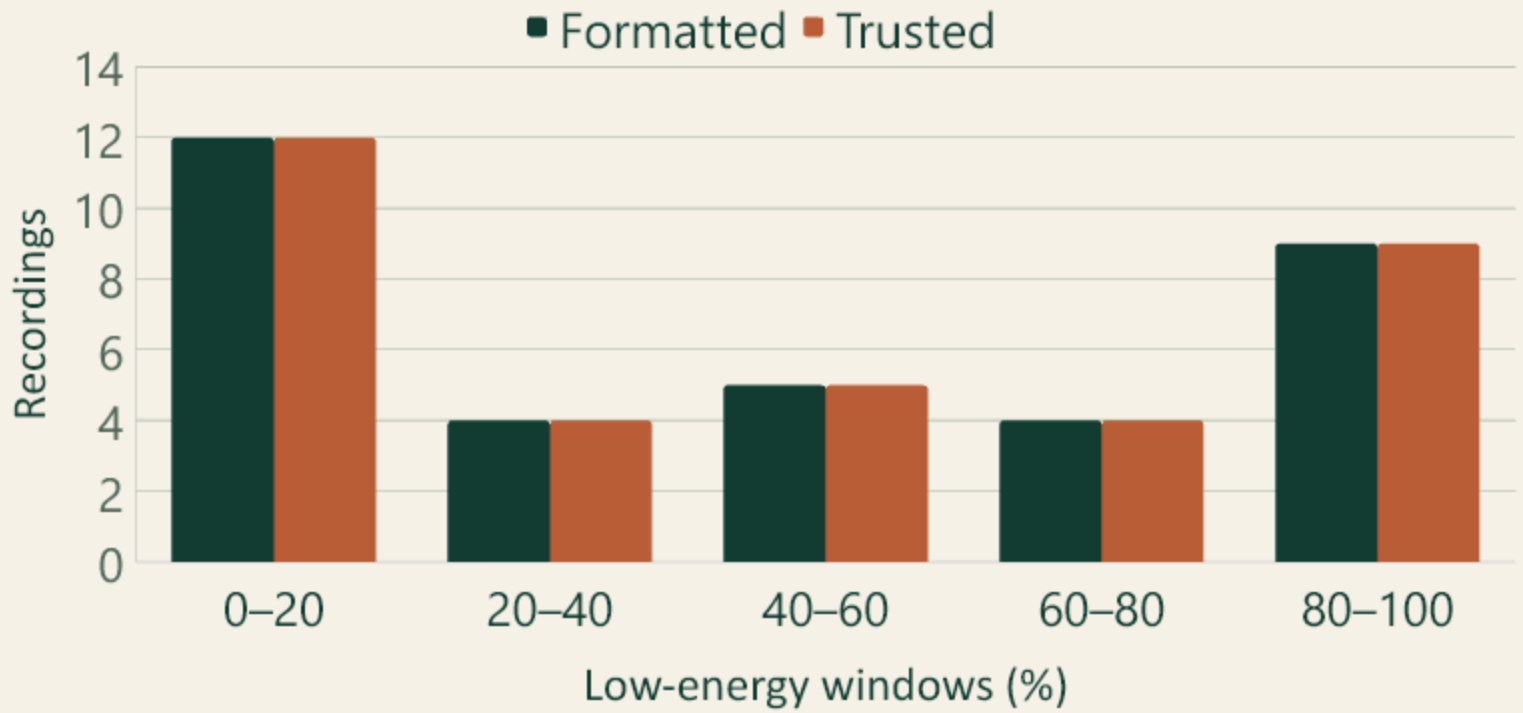
Did filtering change species representation?



100 images before / 100 after / Jensen–Shannon divergence: 0.0 bits

Quiet intervals remain in the evidence

Recordings grouped by low-energy window percentage



9 warnings

34 retained

Complete recordings remain available

50-ms windows below −50 dBFS / flag at ≥80% / same observed distributions

Five collections, 6,021 embeddings

6,021

embeddings
from 198 assets

Image / CLIP	**100**
Audio / CLAP	**357**
Text / E5	**547**
Image + text / CLIP	**1,763**
Audio + text / CLAP	**3,254**

Reuse requires source/model compatibility and exact segment coverage

Three encoders, three relationships



Aligned space: cosine / across models: combine ranks (RRF)

Similarity is an angle, not a probability

$\cos(q, x) = (q \cdot x) / (\|q\| \|x\|)$

Illustrative 2D unit vectors: q = (1, 0) / cosine = dot product

Candidate vector x	Cosine	Relative alignment
(0.96, 0.28)	0.96	Closest of these three
(0.60, 0.80)	0.60	Less aligned
(0.00, 1.00)	0.00	Perpendicular to q

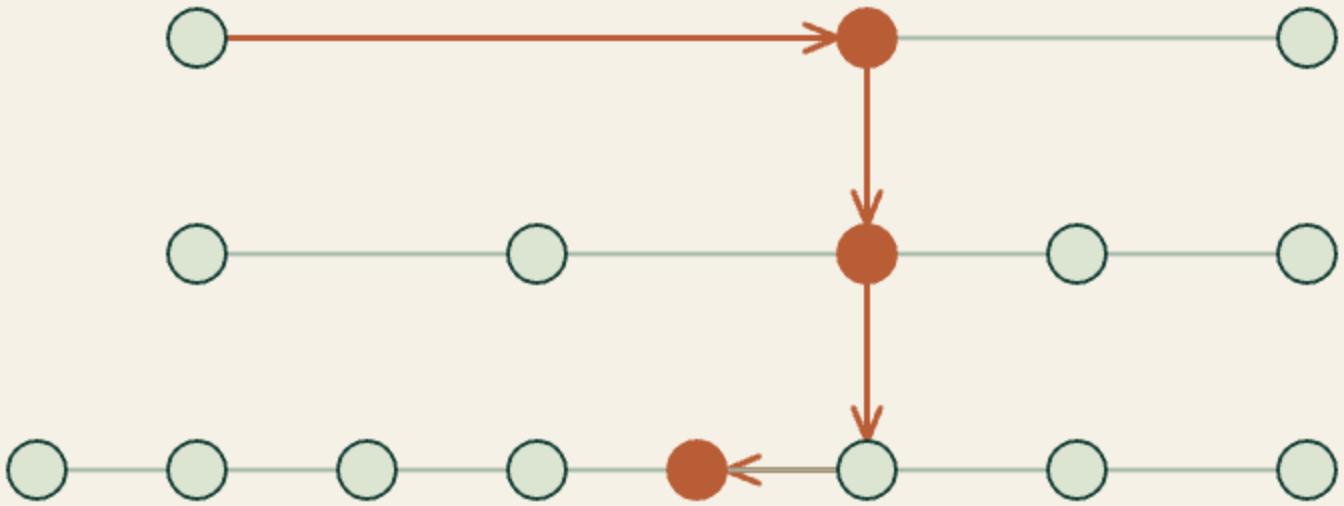
Higher = closer in this space. No universal relevance threshold.

Production: 384 / 512 dimensions. A score of 0.96 is not 96% confidence.

Finding neighbours: HNSW in Chroma

Hierarchical Navigable Small World / approximate nearest neighbours

Sparse



Longer hops

Refine locally

Cosine ranks candidates

Denser

Our exact path: scan all eligible vectors when N <= 64

Schematic graph / ANN may miss a true neighbour / no measured recall claim

Task 1: a real image query

REFERENCE IMAGE



Query / CLIP

SELECTED MATCHES / RANKS 1 AND 3



#1 / cosine 0.912



#3 / cosine 0.871

5 distinct assets / reference excluded / 4 lynx labels and 1 wolf label

Task 2: one query, images and text

“Iberian lynx in its natural habitat”

IMAGE #5 / cosine 0.332



TEXT #4 / cosine 0.778

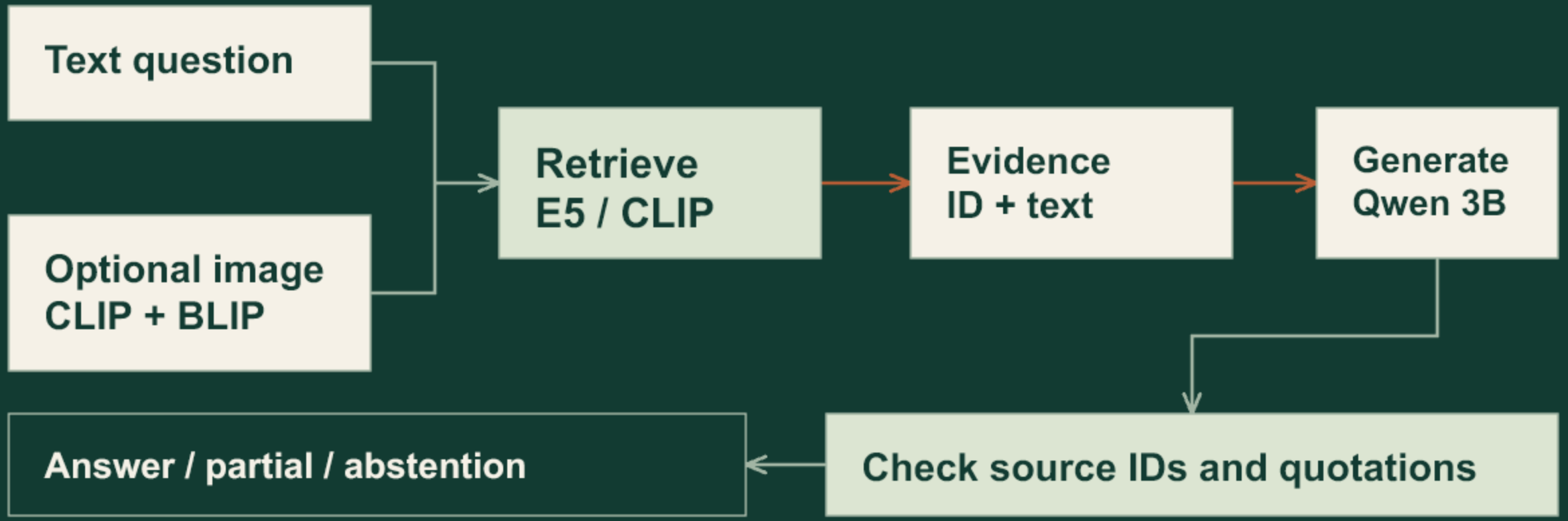
“About half the size of Eurasian Lynx”

Catalogue label: Lynx pardinus

8 results / 4 images + 4 texts / explicit diversity policy

Interface: top-k 1-50 / default 8 / distinct assets with explicit diversity

Task 3: answer from retrieved evidence



Technical errors stay separate

Qwen receives text. Source matches do not prove semantic correctness.

The frog case: evidence and uncertainty

Retrieved candidate

Pelophylax perezi

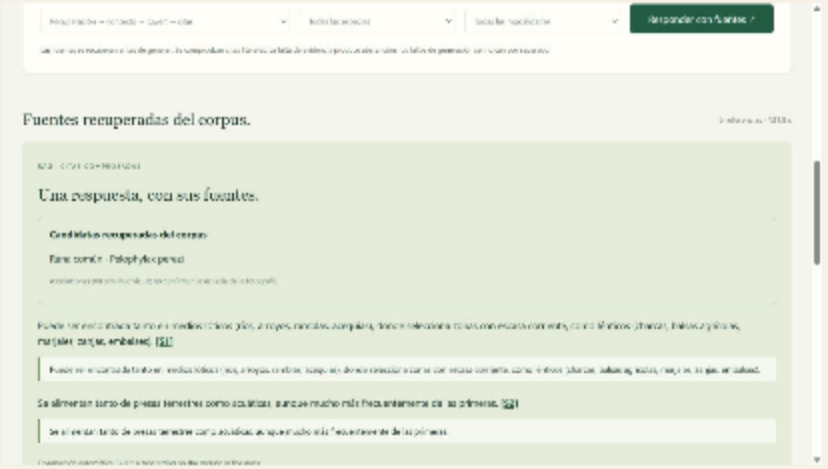
Habitat [S1]

“areas with little current”

Diet [S2]

“terrestrial and aquatic prey”

Actual interface / Spanish



Separate test: complete-genome request / abstained

Real model runs / translated extracts / species identity remains unconfirmed

The second run adds zero new vectors



16 steps / isolated storage / real CPU models

Reconciliation checks outcomes. This synthetic run is not a relevance benchmark.

How we tested the pipeline

Layer	Concrete check	Recorded result
Offline	Generated faults and contract assertions	364 tests passed
Integration	All zones twice, deterministic encoders	1 test passed
CPU CLI	Real decoders and cached models	7 vectors, then 0 new
Browser	Selected top-k reaches HTTP handlers	8 checks passed

Failure assertions inspect stored state and recovery

Copy mismatch, failed completion record, missing vector parts, accepted bytes retained

An input can fail before it reaches storage

lynx.jpg

Harmless MZ header / synthetic fixture

BLOCKED

Before hashing or upload

Corrupt files / empty inputs
Archive expansion / resource limits

Watermark warning retains the image

71.82% combined statement coverage / input screening does not replace antivirus

A query with a traceable answer

A result we can trace back to its source



Next: labelled retrieval and answer evaluation with specialist review